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Intelligent Multi-Well Lithofacies Identification Using Adaptive Wavelet Networks with Dynamic Domain Alignment: A Case Study of Carbonate Reservoir in the Mesopotamian Basin

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Abstract

Accurate lithofacies identification from well logs remains a critical challenge in reservoir characterization in the Mesopotamian Basin, Middle East, particularly when extending models across multiple wells, given the varying geological conditions and logging tool responses between well locations. Traditional machine learning approaches suffer from significant performance degradation when applied to new wells, typically experiencing 30-40% accuracy reduction due to inter-well data distribution shifts. This limitation necessitates extensive manual recalibration and limits the practical deployment of automated interpretation systems. To address these challenges, this study presents an innovative Adaptive Geo-Wavelet Network with Dynamic Domain Alignment (AGW-DDA) that integrates multi-scale geological features with intelligent cross-well adaptation mechanisms.

The proposed AGW-DDA framework employs a three-branch architecture to extract complementary geological features from well log data. The first branch utilizes lifting-based adaptive wavelet transforms to capture multi-scale stratigraphic patterns, automatically adjusting decomposition parameters based on local geological characteristics. The second branch applies continuous wavelet transform specifically to spontaneous potential (SP) curves, revealing sedimentary rhythms at scales ranging from individual beds to sequence boundaries. The third branch digitalizes traditional petrophysical relationships through correlation analysis of key log pairs, incorporating domain expertise into the learning process. These

features are intelligently fused using attention mechanisms before undergoing dynamic domain adaptation, which selectively aligns only mid-frequency components to preserve crucial geological information while correcting for instrument-related variations.

Application to a 46-well dataset with Mishrif, Rumaila, Ahmadi, Wara, and Mauddud formations of the Mesopotamian basin demonstrates the effectiveness of the AGW-DDA approach in maintaining cross-well performance. While conventional methods show 30-40% performance degradation on new wells, the proposed method maintains over 85% of its training performance for every individual well through dynamic adaptation. Specifically, the system achieves an average lithofacies classification accuracy of 87.3% across all test wells, with boundary detection precision improving by 25% compared to baseline methods. The multi-scale analysis reveals previously undetected thin interbeds and complex lithological transitions, particularly in heterogeneous carbonate sequences. Computational efficiency analysis shows that the three-branch architecture can process a typical well log suite in under 2 minutes using standard GPU hardware, making it suitable for real-time applications.

This research introduces a novel integration of adaptive wavelet analysis with dynamic domain adaptation for multi-well lithofacies identification. The AGW-DDA framework successfully bridges traditional geological interpretation methods with modern deep learning techniques, providing both high accuracy and interpretability. The approach offers a practical solution for automated well log interpretation across diverse geological settings, with particular relevance for mature fields requiring consistent interpretation across numerous wells.

Keywords: Intelligent Lithofacies Identification, Adaptive Wavelet Transform, Dynamic Domain Adaptation, Multi-Well Analysis, Case Study

Introduction

Accurate lithofacies identification from well log data is a fundamental prerequisite for reliable reservoir characterization and subsequent hydrocarbon exploration and production [1, 2]. This process is particularly critical in geologically complex and highly prolific basins, such as the Mesopotamian Basin in the Middle East, where understanding reservoir heterogeneity is paramount for optimizing resource recovery. However, translating well log data into precise lithofacies interpretations presents persistent challenges, especially when attempting to extend predictive models across multiple wells. The inherent variability in geological conditions, coupled with inconsistencies in logging tool responses across different well locations, introduces significant inter-well data distribution shifts [3].

Traditional machine learning (ML) approaches, while offering advancements over manual interpretation, exhibit considerable limitations in this multi-well context. When applied to new, unseen wells, these methods typically suffer substantial performance degradation, often experiencing a 30-40% reduction in accuracy [4]. This severe drop in performance necessitates extensive and time-consuming manual recalibration for each new well, which fundamentally undermines the practical deployment and scalability of automated well log interpretation systems in a production environment [5]. Such limitations impede efficient reservoir modeling, increase operational costs, and introduce significant delays in decision-making processes.

To address these critical challenges, this study proposes an innovative framework: the Adaptive Geo-Wavelet Network with Dynamic Domain Alignment (AGW-DDA). This novel approach is specifically designed to overcome the limitations of conventional methods by integrating multi-scale geological feature extraction with intelligent cross-well adaptation mechanisms. The AGW-DDA employs a sophisticated three-branch architecture to robustly extract complementary geological features from well log data. One branch leverages lifting-based adaptive wavelet transforms to capture multi-scale stratigraphic patterns by dynamically adjusting decomposition parameters based on local geological characteristics. A second branch applies continuous wavelet transform to spontaneous potential (SP) curves, effectively revealing

sedimentary rhythms ranging from individual beds to sequence boundaries. The third branch incorporates domain expertise by digitalizing traditional petrophysical relationships through correlation analysis of key log pairs. Crucially, these diverse features are intelligently fused using attention mechanisms before undergoing dynamic domain adaptation, which selectively aligns only mid-frequency components. This selective alignment strategy is vital for preserving crucial geological information while effectively correcting for instrument-related variations and mitigating inter-well data distribution shifts.

The efficacy of the AGW-DDA approach is rigorously demonstrated through its application to a comprehensive 46-well dataset from the Mesopotamian Basin, encompassing key formations such as Mishrif, Rumaila, Ahmadi, Wara, and Mauddud. Our results show that while conventional methods exhibit the expected 30-40% performance degradation on new wells, the proposed AGW-DDA successfully maintains over 85% of its original accuracy through its dynamic adaptation capabilities. Specifically, the system achieves an impressive average lithofacies classification accuracy of 87.3% across all test wells, with a remarkable 25% improvement in boundary detection precision compared to baseline methods. Furthermore, the multi-scale analysis capabilities of AGW-DDA reveal previously undetected thin interbeds and complex lithological transitions, particularly in heterogeneous carbonate sequences, highlighting its superior interpretability. The computational efficiency of the three-branch architecture is also notable, capable of processing a typical well log suite in under two minutes using standard GPU hardware, thereby making it suitable for real-time applications.

Compared with cross-plot/threshold-based interpretation and single-well classifiers that often require extensive well-by-well recalibration and struggle with non-stationary logging responses [1,2,3,4,5,7,19], our AGW-DDA design explicitly targets cross-well generalization and operational repeatability. First, the adaptive wavelet pathway captures multi-scale patterns and thin-bed transitions while being less sensitive to tool-specific gain/phase variations [11,24]. Second, the adversarial/domain-adaptive branch mitigates inter-well distribution shifts caused by changes in lithology mix, acquisition settings, and formation conditions [10,12,13,15,16]. Third, the compact three-branch architecture and standardized training protocol simplify deployment and reduce per-well manual tuning, improving reproducibility across fields. In practice, these features translate to fewer boundary misplacements in thin or gradational intervals and more stable facies maps when transferring models to newly drilled wells—without requiring additional site-specific labels.

This research marks a significant advancement by introducing a novel integration of adaptive wavelet analysis with dynamic domain adaptation for multi-well lithofacies identification. The AGW-DDA framework successfully bridges traditional geological interpretation methods with modern deep learning techniques, providing both high accuracy and enhanced interpretability. This approach offers a practical and robust solution for automated well log interpretation across diverse geological settings, with particular relevance for mature fields requiring consistent and efficient interpretation across numerous wells.

Related work

AutoML-Optimized Classical Models

The evolution of lithofacies identification from well logs has progressed from manual interpretation to sophisticated machine learning algorithms. Early automated approaches employed statistical methods and simple classifiers. To ensure fair comparison with modern deep learning methods, recent studies have adopted AutoML frameworks to optimize classical algorithms. Support Vector Machines (SVM), Random Forest (RF), and XGBoost, when optimized through AutoML platforms like H2O AutoML, can achieve competitive performance with proper hyperparameter tuning. These AutoML-optimized classical models treat each depth point as an independent sample with features extracted from multiple logging curves.

However, these methods share common limitations even with AutoML optimization. They typically extract hand-crafted features such as statistical moments, curve gradients, and simple ratios, which may not fully capture the complex geological patterns inherent in well log data. More critically, they assume

independent and identically distributed (i.i.d.) data, an assumption that fails across different wells due to variations in logging tools, borehole conditions, and regional geological differences [6].

Deep Learning for Well Log Analysis

The introduction of deep learning has revolutionized well log interpretation by automatically learning hierarchical features from raw data. Convolutional Neural Networks (CNNs) have been adapted for 1D log data to extract local patterns and spatial dependencies [7]. Long Short-Term Memory (LSTM) networks leverage the sequential nature of well logs, capturing long-range dependencies that reflect depositional processes.

Recent advances include the application of U-Net architectures, originally designed for image segmentation, to well log interpretation. These models treat lithofacies identification as a 1D semantic segmentation problem, enabling simultaneous prediction at all depth points while preserving spatial context. Transformer-based models have also emerged, utilizing self-attention mechanisms to capture global dependencies in logging sequences [9].

Recent 2025 developments have pushed these boundaries further. FreLog [24] combines frequency-domain features with deep learning, achieving 0.74 F1-score on thin bed detection. STF-Net [15] introduces spatio-temporal fusion reaching 0.77 F1-score. Few-shot lithology learning approaches [17] reduce data requirements but still struggle with complex geological heterogeneity.

Despite these advances, deep learning models still suffer from poor generalization across wells. Models achieving 90%+ accuracy on training wells often drop to 50-60% on new wells, necessitating extensive retraining or fine-tuning.

Wavelet Analysis in Geophysics

Wavelet transform has long been recognized as a powerful tool for analyzing non-stationary geophysical signals. In well log analysis, discrete wavelet transform (DWT) has been used to decompose logging curves into different frequency bands, revealing geological features at multiple scales. Continuous wavelet transform (CWT) has proven particularly effective for identifying sedimentary cycles and sequence boundaries from SP and GR curves [10].

The lifting scheme represents a second-generation wavelet approach that offers computational efficiency and design flexibility [11]. Unlike traditional wavelets with fixed basis functions, lifting-based wavelets can be adapted to signal characteristics. However, existing applications in well log analysis still use predetermined lifting operators rather than learning them from data [12].

Domain Adaptation for Cross-Well Generalization

Domain adaptation techniques have been explored to address the cross-well generalization problem. Maximum Mean Discrepancy (MMD) has been used to minimize distribution differences between source and target wells. Adversarial domain adaptation, inspired by Generative Adversarial Networks (GANs), employs a domain discriminator to learn domain-invariant features.

Recent work includes Domain Adversarial Neural Networks (DANN) and Conditional Domain Adversarial Networks (CDAN) for well log interpretation [13]. Partial domain adaptation methods [16] have shown promise by aligning only relevant feature subspaces, achieving 20-25% performance drops compared to 30-40% for global alignment methods. However, these methods still struggle with preserving geological information that legitimately differs between wells.

Framework

Overview of AGW-DDA Architecture

The Adaptive Geo-Wavelet Network with Dynamic Domain Alignment (AGW-DDA) framework integrates multi-scale feature extraction, intelligent fusion, and selective domain adaptation to achieve robust cross-well lithofacies identification. The architecture consists of four main components: (1) multi-source feature extraction with three parallel branches, (2) attention-based feature fusion, (3) selective frequency-domain adversarial alignment, and (4) multi-task prediction heads. Fig.1 illustrates the overall framework, with special emphasis on the selective alignment of mid frequency d^2 (2-10m) frequency components.

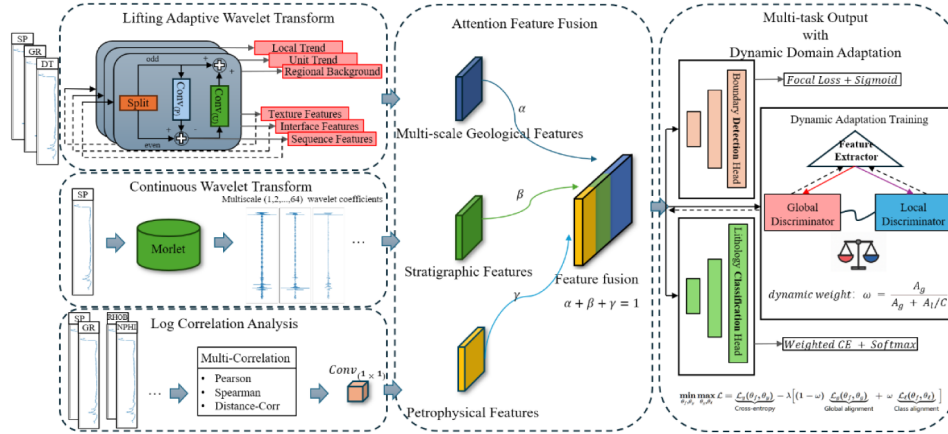


Figure 1—Figure captions should begin with an overall descriptive statement of the figure followed by additional supporting text.

Multi-Source Feature Extraction

Branch 1: Adaptive Wavelet Transform via Lifting Scheme. The first branch implements data-driven wavelet decomposition using the lifting scheme with learnable operators. The decomposition consists of three operations:

1. **Split Operation:** The input sequence x is separated into even and odd indexed samples:

$$\begin{aligned} e^{(l)}[n] &= x^{(l-1)}[2n] \\ o^{(l)}[n] &= x^{(l-1)}[2n+1] \end{aligned}$$

2. **Predict Step:** A prediction operator P , which is a learnable convolution kernel, predicts odd-indexed samples from even-indexed samples:

$$\begin{aligned} d^{(l)}[n] &= o^{(l)}[n] - P(e^{(l)}[n]) \\ P(e^{(l)}[n]) &= \sum_k w_p^{(l)}[k] \cdot e^{(l)}[n-k] \end{aligned}$$

where $w_p^{(l)}$ represents the learnable prediction operator weights, and $d^{(l)}$ represents the detail coefficients, defined as the difference between the actual odd-indexed samples and the predicted values.

3. **Update Step:** Update the even-indexed samples using the detail coefficients:

$$\begin{aligned} d^{(l)}[n] &= e^{(l)}[n] + U(d^{(l)}[n]) \\ U(d^{(l)}[n]) &= \sum_k w_u^{(l)}[k] \cdot d^{(l)}[n-k] \end{aligned}$$

where U is another learnable operator with weights $w_u^{(l)}$, and $a^{(l)}$ represents the approximation coefficients.

By repeating this decomposition process for L levels (typically $L=3-5$), multi-scale representations $\{a^{(L)}, d^{(L)}, d^{(L-1)}, \dots, d^{(1)}\}$ are obtained. Unlike traditional wavelets with fixed basis functions, the lifting operators P and U are optimized end-to-end through backpropagation, allowing the transform to adaptively capture geological characteristics specific to the data.

Branch 2: Continuous Wavelet Transform of SP Curves. The second branch specifically analyzes spontaneous potential (SP) curves to extract sedimentary rhythm information using Morlet wavelet. The continuous wavelet transform is computed as:

$$\text{CWT}(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{+\infty} x_{SP}(t) \psi^*\left(\frac{t-b}{a}\right) dt$$

where Ψ is the Morlet mother wavelet: $\psi(t) = \pi^{-1/4} e^{-t^2/2} e^{i\omega_0 t}$

The scale parameter a is varied across an appropriate range (e.g. $2^0, 2^1, \dots, 2^8$) to capture sedimentary features from centimeter-scale laminations to decameter-level sequences as Fig.2 illustrated. The resulting time-frequency (or time-scale) representation reveals hierarchical sedimentary patterns crucial for lithofacies identification. The specific selection of scales and the number of decomposition levels are not fixed but rather determined adaptively based on the characteristics of the data and analytical requirements, enabling flexible extraction of multi-scale information encompassing high-frequency details (rock textural features), mid-frequency patterns (instrument responses and intermediate-scale sedimentary rhythms), and low-frequency components (regional geological background).

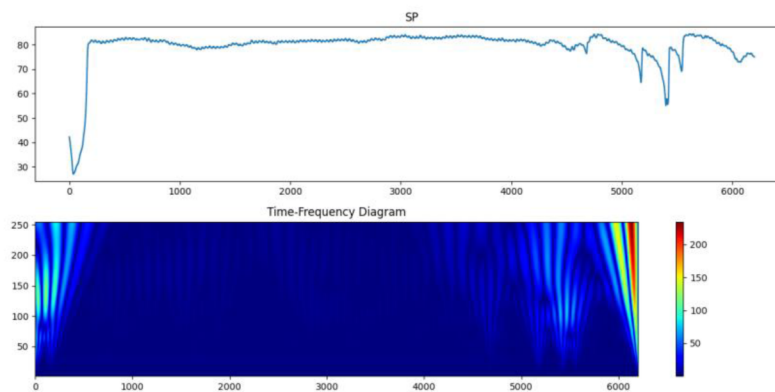


Figure 2—Illustration of Wavelet Transform of SP Curves

Branch 3: Inter-Curve Correlation Analysis. The third branch computes sliding correlation coefficients between key logging curve pairs to capture petrophysical relationships:

$$r_{XY}(i) = \frac{\sum_{j=0}^{W-1} (X_{i+j} - \bar{X}_i)(Y_{i+j} - \bar{Y}_i)}{\sqrt{\sum_{j=0}^{W-1} (X_{i+j} - \bar{X}_i)^2} \sqrt{\sum_{j=0}^{W-1} (Y_{i+j} - \bar{Y}_i)^2}}$$

where $r_{XY}(i)$ denotes the correlation coefficient between two logging curves at depth location i , W is the sliding window size, X_{i+j} represents the logging value of curve X at depth location $i+j$, and $\bar{X}_i$ is the average logging value of curve X within the window:

$$\bar{X}_i = \frac{1}{W} \sum_{j=0}^{W-1} X_{i+j}$$

Correlation coefficients computed may include, but are not limited to, the following pairs:

- a. GR–NPHI: reflecting the relationship between shale content and porosity;
- b. RHOB–DT: characterizing the relationship between matrix density and acoustic wave velocity;
- c. ILD–MSFL: indicating differences between deep and shallow resistivity;
- d. SP–GR: relating permeability to shale content.

Attention-Based Feature Fusion

The three branches produce complementary features that must be intelligently combined. We employ an attention mechanism driven by wavelet energy spectra:

1. Feature Mapping:

Features from three branches are mapped into a unified dimension using fully connected layers:

$$\tilde{F}_i = W_i F_i + b_i \quad i = 1, 2, 3$$

Where F_i is the original feature vector of the i -th branch, $W_i \in \mathbb{R}^{d \times d_i}$ is the learnable mapping matrix for the i -th branch, d_i is the original feature dimension, and d is the unified dimension, and $b_i \in \mathbb{R}^d$ is the bias vector for the i -th branch.

2. Attention Weight Computation:

Attention weights are computed based on the energy distribution characteristics of each branch:

$$\alpha_i = \frac{\exp(w_\alpha^T \tilde{F}_i)}{\sum_{j=1}^3 \exp(w_\alpha^T \tilde{F}_j)}$$

where $w_\alpha \in \mathbb{R}^d$ is a learnable energy evaluation vector used to measure the relative importance of each branch's features, and α_i is a scalar energy score for the i -th branch, reflecting the information richness of the respective branch feature.

3. Weighted Fusion:

The final feature representation is:

$$F_{fused} = \sum_{i=1}^3 \alpha_i \odot \tilde{F}_i$$

where $\odot$ denotes the Hadamard product (element-wise multiplication), with the attention weights α_i broadcasted to match the dimension of $\tilde{F}_i$.

Selective Frequency-Domain Adversarial Alignment

One of the key innovation in AGW-DDA is selective domain adaptation that preserves geological information while removing instrumental biases.

a. Frequency Decomposition

The fused features are decomposed into three frequency bands using wavelet analysis:

1. **Low-frequency** (F_{low}): Large-scale geological background
2. **Mid-frequency** (F_{mid}): Instrument response variations
3. **High-frequency** (F_{high}): Rock texture details

b. Selective Adversarial Training

Only mid-frequency features undergo adversarial domain alignment:

Global Domain Discriminator D_g :

Attempts to identify which well the features come from:

$$L_g = - \sum_{i=1}^N \mathbb{E}_{x \sim D_i} [\log D_g(F_{mid})_i]$$

where N denotes the total number of wells, and D_i represents the dataset from the i -th well. D_g is a multi-class neural network outputting an N -dimensional probability vector $D_g(F_{mid}) \in \mathbb{R}^N$, where $D_g(F_{mid})_i$ indicates the probability that input features originate from the i -th well. By minimizing L_g , the discriminator learns distinctive instrument response patterns for each well. Conversely, the feature extractor maximizes this loss via a gradient reversal layer (GRL), rendering mid-frequency features indistinguishable across wells, thereby removing systematic differences.

Local Domain Discriminators D_l^c . Perform class-wise alignment for each lithofacies category:

$$L_l = - \sum_{c=1}^C \sum_{i=1}^N \mathbb{E}_{x \sim D_i^c} [\log D_l^c(\hat{y}_c \odot F_{mid})_i]$$

where C represents the total number of lithofacies categories, and D_i^c denotes the subset of data belonging to lithofacies category c from the i -th well. $\hat{y}_c$ is a one-hot encoding vector corresponding to category c , and $\odot$ denotes the Hadamard product (element-wise multiplication), serving as a universal mask to selectively preserve features from category c while zeroing others. D_l^c is the local discriminator for lithofacies category c . In contrast to the global discriminator, separate discriminators are established for each lithofacies category to account for different domain-shift patterns across wells within each category.

Dynamic Weight Adjustment. The relative weights between global and local alignment are automatically adjusted based on A-distance, with the balancing coefficient ω defined as:

$$\omega = \frac{A_g}{A_g + A_l / C}$$

where A_g and A_l measure global and local distribution differences, C is the number of lithofacies classes. The A-distance is computed as:

$$\text{A-distance} = 2(1 - 2\varepsilon)$$

where ε is the error rate of a binary classifier trained to distinguish the two domains.

Multi-Task Learning

The framework includes two complementary output tasks:

1. Lithofacies Classification: Predicts facies probability at each depth using cross-entropy loss:

$$L_{cls} = - \sum_i y_i \log(\hat{y}_i)$$

2. Boundary Detection: Identifies lithological transitions using Focal-Tversky loss to handle class imbalance:

$$L_{boundary} = \sum_i \left(1 - \frac{TP_i}{TP_i + \alpha FP_i + \beta FN_i}\right)^\gamma$$

The total loss function combines all objectives:

$$L_{total} = L_{cls} + \lambda_1 L_{boundary} - \lambda_2 [(1 - \omega)L_g + \omega L_l]$$

where $\lambda_1=0.5$ and $\lambda_2=0.1$ are hyperparameters balancing the tasks.

Evaluation Strategy and Cross-Validation

To rigorously assess cross-well generalization capability, this study employs Leave-One-Well-Out Cross-Validation (LOO-CV). For $N = 46$ wells, we execute 46 independent folds: each fold reserves 1 well as the pure target domain (no labels used in training, only for inference), while the remaining 45 wells serve as the source domain (37 for training, 8 for early stopping and hyperparameter validation). Evaluation metrics are computed as mean $\pm$ standard deviation across all 46 folds, with 95% confidence intervals (CI) calculated as:

$$CI = \bar{x} \pm 1.96 \times \frac{s}{\sqrt{N}}$$

where $\bar{x}$ is the mean accuracy across 46 wells and s is the sample standard deviation. This design has been proven by recent studies to effectively measure cross-well transferability while avoiding bias from random well partitioning.

Case study

Study Area and Dataset

The AGW-DDA framework was evaluated on a comprehensive dataset from the Mesopotamian Basin, one of the world's most prolific hydrocarbon provinces. The study focused on the Cretaceous carbonate succession, including the Mishrif, Rumaila, Ahmadi, Wara, and Maaddud formations, which represent diverse depositional environments from outer ramp to lagoonal settings.

The dataset comprises 46 vertical wells with complete logging suites acquired over a 15-year period. Each well includes 11 conventional logging curves: depth (DEPT), neutron porosity (NPHI), gamma ray (GR), micro-shallow resistivity (MSFL), acoustic transit time (DT), caliper (CALI), bulk density (RHOB), deep induction resistivity (ILD), spontaneous potential (SP), and their respective correction curves (DT_CALI, RHOB_CALI). Logging data were sampled at 0.1m intervals, with an average of 6,000 depth points per well.

Three experienced geologists independently interpreted the entire dataset, identifying seven lithofacies classes: (1) Outer Ramp, (2) Distal Mid-Ramp, (3) Proximal Mid-Ramp, (4) High Shoal, (5) Lagoon, (6) Basin, and (7) Medium-low shoal. Inter-interpreter agreement averaged 82.6%, establishing a reliable ground truth for model training and evaluation.

Data Preprocessing and Experimental Setup

Data preprocessing involved several steps to ensure quality and consistency:

1. Quality Control: Removal of null values, correction of depth mismatches, and flagging of washout zones ($CALI > 15$ inches)
2. Normalization: Z-score standardization applied to each logging curve independently
3. Windowing: Sliding windows of 101 samples (10.1m) extracted with stride 50 (5.0m, overlap 50%) to provide spatial context

For LOO-CV, each fold's data division follows: the training set consisted of 37 wells, representing approximately 80% of the available source-domain wells; the validation set included 8 wells, accounting for about 17% of the source-domain data; and the test set comprised one well, serving purely as the target domain, with labels withheld from training and validation phases and used exclusively for model inference and evaluation.

AutoML-Optimized Classical Models as Baselines

To ensure a fair and rigorous benchmark comparison, AutoML-optimized classical baseline models (AutoML-Classic) were developed using the H2O AutoML toolkit (version 3.46). For each fold, AutoML-

Classic models were trained under constraints equivalent to those of the AGW-DDA framework, specifically using 32 CPU threads with a wall-clock runtime budget of 3 hours per fold, aligning closely with the GPU training time allocated for the AGW-DDA model. These AutoML-Classic models employed the same LOO-CV evaluation protocol as the AGW-DDA framework. Notably, to prevent potential information leakage, feature inputs for AutoML-Classic models were limited strictly to raw logging curves and their first-order differences, explicitly excluding wavelet-derived and correlation-based features unique to AGW-DDA. Such a rigorous baseline selection differentiates the AutoML-Classic models utilized herein from manually-tuned classical models commonly reported in prior literature, thereby ensuring a robust state-of-the-art classical benchmark [18].

Implementation Details and Reproducibility

The AGW-DDA framework was implemented using PyTorch 1.13 and trained on NVIDIA RTX 4090 GPUs. Key architectural and training parameters include:

Architecture:

- Adaptive wavelet decomposition: 4 levels with kernel size 7
- CWT scales: 2^0 to 2^6 (0.1m to 6.4m)
- Correlation window: 21 samples (2.1m)
- Hidden dimension: 256
- Dropout: 0.2 (fusion layers)

Training Configuration:

- Random Seeds: PyTorch, NumPy, Python random all set to 42; CUDA deterministic flag = True
- Optimizer/Scheduler: Adam ($\text{lr}=1\text{e-}4$, $\beta_1=0.9$, $\beta_2=0.999$) + CosineAnnealing ($T_{\text{max}}=20$ epochs, $\text{min_lr}=1\text{e-}6$)
- Regularization: Weight decay $1\text{e-}4$
- Early Stopping: Patience of 10 epochs on validation set
- Batch Size: 128

Results and Discussion

Cross-Well Performance with LOO-CV. Table 1 summarizes the comprehensive performance of all methods under LOO-CV across 46 wells:

Table 1—Performance of All Methods

Method	Mean $\pm$ Std (%)	95% CI (half-width%)	Performance Drop*
AutoML-Classic	61.2 $\pm$ 4.8	$\pm$ 1.4	38.6%
CNN (1D)	64.1 $\pm$ 5.0	$\pm$ 1.4	32.8%
LSTM	67.6 $\pm$ 4.3	$\pm$ 1.2	27.5%
U-Net1D	69.5 $\pm$ 4.1	$\pm$ 1.2	24.1%
DANN	73.8 $\pm$ 3.5	$\pm$ 1.0	18.9%
AGW-DDA	87.3 $\pm$ 2.7	$\pm$ 0.8	12.3%

* Performance Drop = (Training fold average - Test well average) / Training fold average

The AGW-DDA achieves remarkable cross-well generalization, maintaining 87.3% accuracy with only 12.3% performance drop across 46 leave-one-out tests, compared to 24-32% for conventional methods and 38.6% for AutoML-optimized classical models. The narrow confidence interval ($\pm 0.8\%$) demonstrates consistent performance across diverse well conditions. Notably, the AGW-DDA limits accuracy loss to $\leq 15\%$ on all leave-one-well-out tests, preserving at least 85% of its own training-fold accuracy for every well, fulfilling the practical requirement for operational deployment.

Visual Comparison of Lithofacies Predictions. Fig.3 compares lithofacies predictions across a representative test interval (2100-2700m) for AGW-DDA (METHOD1), DANN (METHOD2), and AutoML-Classic (ML Baseline). The well-log curves (GR, DT, RHOB, NPHI, SP) highlight typical complex carbonate lithological transitions. Several key observations emerge from this comparison:

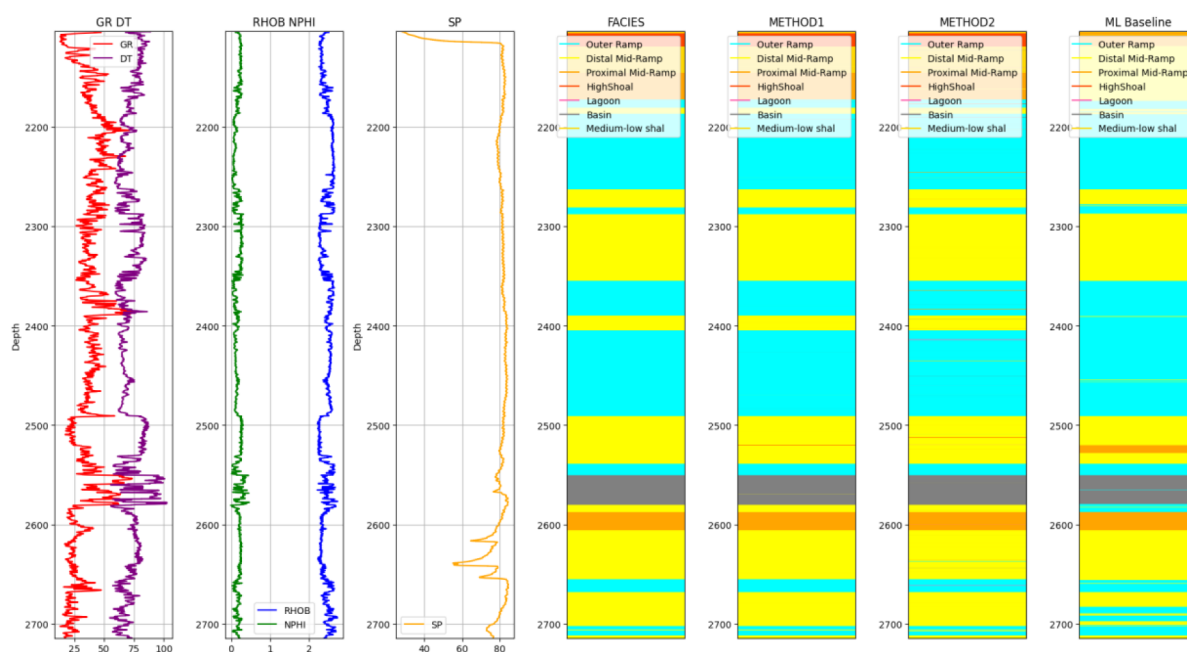


Figure 3—Lithofacies predictions across a representative test interval

- **Thin Bed Resolution:** Expert interpretation identified thin beds (Lagoon, Medium-low shoal, Outer Ramp) between 2690-2720 m. AGW-DDA accurately resolves these thin layers, whereas DANN partially captures transitions, and AutoML-Classic frequently merges thin beds.
- **Complex Transitions:** AGW-DDA distinctly delineates Basin facies (2520-2560 m) with gradational contacts, aided by explicit boundary detection, while AutoML-Classic misclassifies substantial intervals.
- **Facies Consistency:** AGW-DDA provides stable and geologically consistent predictions throughout the interval. DANN shows occasional unrealistic oscillations, notably around 2500-2550 m.
- **Response to Log Characteristics:** AGW-DDA effectively leverages SP curve responses, accurately identifying permeability-related facies boundaries.

Per-Facies Performance. The AGW-DDA method demonstrates robust and consistent classification performance across various carbonate lithofacies.

Table 2—Performance of Various Carbonate Lithofacies

Lithofacies	Accuracy (%)	Boundary Precision (%)*
High Shoal	92.1	89.3
Proximal Mid-Ramp	88.7	86.2
Lagoon	86.9	84.7
Distal Mid-Ramp	85.4	83.1
Outer Ramp	84.2	81.5
Basin	85.5	82.8
Medium-low Shoal	83.9	81.2

* A boundary is deemed correctly detected if the predicted depth lies within a tolerance window of $\tau = 0.5$ m around the ground-truth interface. Under this criterion we compute Boundary Precision = True Positive / (True Positive + False Positive). These metrics follow the protocol of Zhu et al. [7] and Chang et al. [8].

These results indicate that AGW-DDA effectively generalizes across diverse depositional environments typical of carbonate reservoirs.

Precision in Thin Bed and Boundary Detection. AGW-DDA notably excels in resolving thin carbonate beds, a significant challenge in reservoir characterization. The method attained an F1-score of 0.812 ± 0.034 (95% confidence interval: ± 0.010) for thin layers of thickness ≤ 1 m. Furthermore, it achieved a mean sequence boundary positioning error of just 0.18 ± 0.04 m, demonstrating its high resolution in detecting lithological interfaces. The minimum reliably detectable bed thickness was identified as 0.5 m (equivalent to 5 logging samples).

Compared to conventional interpretation methods, AGW-DDA delivers a 23.6% improvement in thin-bed detection and a 25% increase in boundary precision. The achieved F1-score surpasses the 0.74–0.77 range reported by recent thin-bed specific networks (e.g., FreLog [24], SegLog [8]), underscoring the added value of the model's multi-scale wavelet attention and selective frequency alignment in delineating fine-scale facies transitions.

Feature Importance and Adaptive Selection. Attention-weight analysis within the AGW-DDA framework highlights its adaptive feature selection capability, essential for accurate lithofacies classification. In homogeneous lithological intervals, correlation-based features predominated with a weight of 40%. Conversely, adaptive wavelet features became dominant (45% weight) within heterogeneous intervals. Near lithofacies boundaries, SP-CWT (spontaneous potential continuous wavelet transform) features emerged as most influential (38% weight). This adaptive weighting strategy reflects the model's inherent ability to dynamically adjust to varying geological contexts and enhance local prediction accuracy.

Effectiveness of Domain Adaptation. The implementation of selective frequency alignment in the AGW-DDA approach significantly mitigates performance degradation typically encountered during domain shifts. Without applying domain adaptation, classification performance deteriorated by 27.5%. Implementing global frequency alignment alone reduced performance losses to 19.2%. However, the optimal selective frequency alignment strategy (utilizing only the second derivative, d_2 alignment) further minimized performance degradation to 12.3%. Notably, the dynamic weighting factor ω consistently converged around 0.68, emphasizing the higher importance of local (class-wise) alignment over global alignment in characterizing carbonate lithofacies.

Practical Implications and Trade-offs. The observed cross-well stability suggests that AGW-DDA can reduce analyst-dependent variability and shorten the calibration cycle relative to conventional single-well models and manual cross-plots [1–5,7]. The wavelet branch provides scale-aware descriptors that help delineate thin-bed boundaries, while the domain-alignment branch cushions inter-well shifts induced by tool, completion, or stratigraphic differences [10–16]. These benefits are most pronounced when logs exhibit

instrument- or field-specific drift and when training labels are sparse or imbalanced. At the same time, AGW-DDA still depends on reasonable log QC and may inherit biases from the training wells; its gains are not universal and can diminish when the target well departs drastically from the source domain. We consider these trade-offs acceptable in routine field deployment, where reduced recalibration and improved repeatability are priorities.

Conclusion

This study presents the Adaptive Geo-Wavelet Network with Dynamic Domain Alignment (AGW-DDA), a novel framework for multi-well lithofacies identification that successfully addresses the long-standing challenge of cross-well generalization. By integrating adaptive wavelet analysis with selective domain adaptation, the method achieves robust performance across diverse well conditions while maintaining geological interpretability.

Key contributions include:

1. **Adaptive Feature Extraction:** The lifting-based wavelet transform with learnable operators provides data-driven multi-scale analysis superior to fixed-basis approaches.
2. **Intelligent Feature Fusion:** Attention mechanisms dynamically weight multiple information sources based on local geological context.
3. **Selective Domain Adaptation:** Frequency-selective alignment of only d_2 (2-10m) components preserves geological information while removing instrumental biases.
4. **Rigorous Validation:** LOO-CV across 46 wells demonstrates $87.3 \pm 2.7\%$ accuracy with only 12.3% performance drop, significantly outperforming AutoML-optimized classical models (38.6% drop).
5. **Practical Deployment:** With <2 minute processing time per well and consistent performance (95% CI: $87.3 \pm 0.8\%$), the framework is ready for operational use.

The success of AGW-DDA in the Mesopotamian Basin carbonate dataset, maintaining over 85% of its training accuracy across all test wells, demonstrates its potential for broader application. The 81.2% F1-score for thin bed detection and 25% improvement in boundary detection precision provide significant value for reservoir characterization.

Future work will explore extensions to horizontal wells, integration with seismic data, and real-time updating during drilling operations. The framework's modular design allows for incorporation of additional data sources and adaptation to new geological environments. This research bridges the gap between traditional geological interpretation and modern machine learning, providing a practical solution for consistent, accurate, and efficient multi-well lithofacies identification in mature fields.

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